Introduction to Python for Basic Calculations

1. Introduction

In this practical, you will use a computer programming language called Python to perform some basic calculations, similar to calculations you could perform on a calculator or in Excel. You will do this by working through an interactive notebook exercise that combines the instructions for this class with snippets of computer code that you will executed in order as you work through the exercises.

This laboratory session is **not** designed to teach you how to be a computer programmer! That would take a little longer than a three-hour class. Rather you will **use and modify** code that is already written and tested in order to better understand how calculations are performed. Hopefully you will also get an appreciation for how powerful computer programming can be in helping us perform quantitative analysis.

1.1. Intended Learning Outcomes

In the Python notebook you will download (instructions below) are step-by-step instructions to introduce the Python environment and perform some calculations. By the end of this practical, each student will be able to:

- Run existing blocks of Python code in an IPython environment
- Understand how to conduct basic calculations in Python
- Work out units of measurement for calculations of forces and pressures
- Understand how scientific numbers are represented in scientific computing

2. Download and Run the IPython Notebook

2.1. Download the notebook

The notebook file **Python_Intro_Basic_Calculations.ipynb** can be found on Moodle. Download this file and unzip it to a new folder in your workspace where you usually store your coursework.

2.2. Open the notebook

You **cannot** open this file by double clicking on it. Instead we will open the notebook with some software called Jupyter Notebook. The Jupyter notebook is an extension of your internet browser so will open as a new tab inside Chrome/Firefox/Edge/Safari. To open Jupyter Notebook:

- *Launch **jupyter notebook** from the start menu within the Anaconda folder*
- *Navigate to the folder with the **Python_Intro_Basic_Calculations.ipynb** and click on it to open it.*
- *Follow the instructions inside the notebook once opened.*